SOFTWARE REQUIREMENTS SPECIFICATION

IEEE 830 FORMAT

Festival Explorer

Discover the Festivals of India □□

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An Interactive Web Application for Exploring
the Cultural and Festive Heritage of India
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Table of Contents

Document Revision History

Version	Date	Author	Description of Change
0.1	01 Aug 2026	Project Team	Initial draft – scope and feature list
0.5	08 Aug 2026	Project Team	Added functional & non-functional requirements
1.0	18 Aug 2026	Project Team	Final review, formatting, and submission version

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, unambiguous, and detailed description of the functional and non-functional requirements of the "Festival Explorer – Discover the Festivals of India" web application. This document is intended to serve as the single source of truth for the design, development, testing, and evaluation of the system. It defines what the system will do, the constraints under which it will operate, and the quality attributes it must satisfy.

This document follows the IEEE 830-1998 recommended practice for Software Requirements Specifications and is written to be useful to students, project guides, evaluators, and any future developer who extends the system.

1.2 Document Conventions

- Requirements are uniquely identified using the format FR-XX (Functional Requirement) and NFR-XX (Non-Functional Requirement).
- Priority levels used: High (must be implemented for the system to be usable), Medium (important but not blocking), Low (enhancement / stretch goal).
- The terms "system", "application", and "Festival Explorer" are used interchangeably to refer to the software being specified.
- The term "user" refers to any person interacting with the system unless a specific user class is named.

1.3 Intended Audience and Reading Suggestions

Audience	Suggested Use of this Document
Faculty / Project Guide / Evaluator	Sections 1, 2, 3, 5, 8, 9, 10 for scope, requirements, and evaluation criteria
Student Development Team	Entire document, especially Sections 3, 4, 6, 7 for implementation
UI/UX Designer	Sections 3, 4.1, 6, and Appendix A for screen and interaction design
Backend / Database Developer	Sections 6, 7 for architecture and data model
Software Tester	Sections 3, 5, 9, 10 for test case and acceptance criteria design

1.4 Project Scope

Festival Explorer is a web-based educational and cultural discovery platform that allows users to explore the diverse festivals celebrated across India. The system organizes festival information by state, by month, and by category (religious, harvest, seasonal, regional, national), and presents details such as history, significance, rituals, traditional food, and attire for each festival.

The platform includes an interactive clickable map of India, a festival calendar view, a simple recommendation feature based on user-selected preferences, and short quizzes to test and reinforce cultural knowledge. The system is designed as a content-driven informational and educational application rather than a transactional or e-commerce platform, keeping the scope realistic and achievable for a student team within one academic semester.

In Scope

- Browsing festivals by state, month, and category
- Detailed festival information pages (history, rituals, food, attire, images)
- Interactive India map with state-wise festival markers
- Festival calendar (monthly view)
- User registration, login, and profile-based preferences
- Personalized festival recommendations based on selected interests/region
- Cultural quizzes with scoring
- Search and filter functionality
- Favorites / bookmarking of festivals
- Admin panel for content management (add/edit/delete festival data)

Out of Scope (for the current version)

- Online ticket booking or e-commerce transactions
- Real-time chat, social networking, or user-to-user messaging
- Native mobile applications (Android/iOS) – the system is web-only and responsive
- Multi-language / regional language translation (proposed as a future enhancement)
- Payment gateway integration

1.5 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- Ministry of Tourism, Government of India – publicly available information on Indian festivals.
- Ministry of Culture, Government of India – publicly available cultural heritage resources.
- React.js Official Documentation – https://react.dev
- Node.js Official Documentation – https://nodejs.org
- MongoDB Official Documentation – https://www.mongodb.com/docs

1.6 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
UI / UX	User Interface / User Experience
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
JWT	JSON Web Token, used for authentication
Admin	A user with elevated privileges to manage site content
Visitor	An unregistered/guest user browsing the public site
Recommendation Engine	A rules-based module suggesting festivals based on user-selected preferences

Term	Definition
Responsive Design	A design approach ensuring the UI adapts to different screen sizes

2. Overall Description

2.1 Product Perspective

Festival Explorer is a new, self-contained web application. It is not a modification of an existing system. The application follows a standard three-tier architecture consisting of a presentation layer (frontend web client), an application/business logic layer (backend REST API), and a data layer (database). It is designed to run independently and does not depend on any external institutional system, though it may optionally integrate with public map and calendar libraries.

2.2 Product Functions (Summary)

The major functions provided by the system are summarized below; each is elaborated in Section 3 as a distinct system feature.

1. User Account Management – registration, login, logout, and profile/preference management.
2. Festival Browsing – view festivals by state, month, or category.
3. Festival Detail View – view in-depth cultural information for a selected festival.
4. Interactive India Map – explore festivals visually through a clickable state-wise map.
5. Festival Calendar – view festivals arranged in a monthly calendar layout.
6. Personalized Recommendations – receive suggested festivals based on stated preferences.
7. Cultural Quiz Module – attempt quizzes and view scores/feedback.
8. Search and Filter – quickly locate festivals using keywords and filters.
9. Favorites / Bookmarks – save festivals of interest for quick future access.
10. Admin Content Management – add, edit, and remove festival records and quiz content.

2.3 User Classes and Characteristics

User Class	Description	Technical Expertise
Visitor (Guest)	Any unregistered user who can browse festivals, use the map, calendar, and search, but cannot save favorites, get personalized recommendations, or take quizzes for scored history.	Low – general public
Registered User	A user who has created an account; can bookmark favorites, set preferences for recommendations, and track quiz scores/history.	Low to Medium
Administrator	Manages the master data of the platform — festivals, states, categories, and quiz questions — through a secured admin panel.	Medium – comfortable with web forms/dashboards

2.4 Operating Environment

- Client Side: Any modern web browser – Google Chrome, Mozilla Firefox, Microsoft Edge, Safari (latest two major versions).

- Server Side: Node.js runtime environment hosted on a cloud platform (e.g., Render, Railway, or an institutional server).
- Database: MongoDB (NoSQL) or MySQL (relational), depending on final team decision — see Section 6.
- Devices: Desktop, laptop, tablet, and mobile devices via a responsive web layout; no separate native app required.

2.5 Design and Implementation Constraints

- The system must be developed using open-source / free-tier technologies suitable for a student budget.
- The project timeline is limited to one academic semester; therefore, the scope is intentionally kept to a single, well-defined web platform.
- The application must be responsive and function correctly on both desktop and mobile browsers.
- Festival content (text and images) must be sourced from publicly available, non-copyrighted, or properly credited material.
- The system must be built using a layered architecture to keep frontend, backend, and database concerns clearly separated for ease of evaluation and future extension.

2.6 Assumptions and Dependencies

- Users have access to a device with an internet connection and a modern web browser.
- Festival data (history, dates, rituals) is relatively static and updated periodically by the Admin rather than in real time.
- Third-party libraries used for the interactive map and calendar (e.g., an SVG India map library, a calendar UI library) remain freely available under permissive licenses.
- The project is evaluated in a controlled academic environment and is not expected to handle production-scale concurrent traffic.

3. System Features (Functional Requirements)

This section describes each major feature of Festival Explorer along with its associated functional requirements, priority, and basic flow. Each functional requirement is uniquely numbered for traceability (see Section 9).

3.1 User Registration, Login, and Profile Management

Description: Allows a person to create an account, log in securely, and manage basic profile information and cultural preferences (favorite states/categories) used for recommendations.

Priority: High

ID	Requirement
FR-01	The system shall allow a new user to register using name, email, and password.
FR-02	The system shall validate that the email is unique and properly formatted before registration.
FR-03	The system shall allow a registered user to log in using email and password.
FR-04	The system shall securely hash and store user passwords (never in plain text).
FR-05	The system shall allow a logged-in user to update profile details and cultural preferences.
FR-06	The system shall allow a user to log out, ending the active session.
FR-07	The system shall allow a user to reset a forgotten password via a registered email.

3.2 Browse Festivals (State-wise, Month-wise, Category-wise)

Description: The core browsing feature allowing any visitor to explore the festival catalog using three intuitive dimensions.

Priority: High

ID	Requirement
FR-08	The system shall display a list of Indian states/union territories; selecting one shall show festivals celebrated there.
FR-09	The system shall display a list of months; selecting one shall show festivals celebrated in that month.
FR-10	The system shall allow filtering festivals by category (e.g., Religious, Harvest, Seasonal, National, Regional).
FR-11	The system shall display festival results as a grid/list of cards showing name, state, date range, and a representative image.
FR-12	The system shall allow combining more than one filter at a time (e.g., state + category).

3.3 Festival Detail Page

Description: A dedicated page presenting the complete cultural profile of a selected festival.

Priority: High

ID	Requirement
FR-13	The system shall display the festival name, associated state(s), category, and celebration dates/month.
FR-14	The system shall display the historical background and cultural significance of the festival.
FR-15	The system shall display associated rituals and traditions.
FR-16	The system shall display traditional food associated with the festival.
FR-17	The system shall display traditional attire associated with the festival.
FR-18	The system shall display a photo gallery of relevant images for the festival.
FR-19	The system shall provide a button for a logged-in user to add/remove the festival from Favorites.

3.4 Interactive India Map

Description: A visual, clickable map of India where each state is a selectable region that reveals festivals celebrated there.

Priority: High

ID	Requirement
FR-20	The system shall render an SVG/vector map of India with all states and union territories selectable.
FR-21	The system shall highlight a state on mouse hover/tap with its name and festival count.
FR-22	The system shall, upon clicking/tapping a state, display a panel or navigate to a page listing festivals of that state.
FR-23	The system shall visually indicate (e.g., shading intensity) states with a higher number of documented festivals.

3.5 Festival Calendar

Description: A monthly calendar view that plots festivals on their respective dates so users can plan around them.

Priority: Medium

ID	Requirement
FR-24	The system shall display a calendar view for the current month by default, with navigation to other months/years.
FR-25	The system shall mark dates that have one or more associated festivals.
FR-26	The system shall, upon clicking a marked date, show the name(s) of festival(s) with a link to their detail page.

3.6 Personalized Festival Recommendations

Description: A lightweight, rules-based recommendation feature that suggests festivals to a registered user based on preferences such as favorite states, categories, or previously favorited festivals.

Priority: Medium

ID	Requirement
FR-27	The system shall allow a registered user to select preferred states and/or categories during onboarding or from profile settings.
FR-28	The system shall generate a “Recommended for You” list based on the user's stated preferences and favorites.
FR-29	The system shall update recommendations when the user changes preferences or adds new favorites.

Implementation Note: For the academic scope of this project, the recommendation logic is a rules-based/weighted scoring approach (matching preferred states/categories) rather than a machine-learning model, keeping the feature realistic to implement and demonstrate.

3.7 Cultural Quiz Module

Description: Short multiple-choice quizzes on Indian festivals to make cultural learning engaging.

Priority: Medium

ID	Requirement
FR-30	The system shall present multiple-choice quiz questions related to festivals, categorized by state or theme.
FR-31	The system shall evaluate submitted answers and display the score immediately upon completion.
FR-32	The system shall show the correct answer and a short explanation for each question after submission.
FR-33	The system shall store quiz history/scores for a logged-in user for later viewing.

3.8 Search and Filter

Description: A global search feature to quickly locate festivals by name or keyword.

Priority: Medium

ID	Requirement
FR-34	The system shall provide a search bar accessible from the main navigation.
FR-35	The system shall return matching festivals based on festival name, state, or keyword in description.
FR-36	The system shall display “no results found” with a suggestion to adjust search terms when no match exists.

3.9 Favorites / Bookmarks

Description: Allows registered users to save festivals for quick access later.

Priority: Low

ID	Requirement
FR-37	The system shall allow a registered user to bookmark or remove a festival as a favorite.
FR-38	The system shall display a dedicated “My Favorites” page listing all bookmarked festivals.

3.10 Admin Panel – Content Management

Description: A secured, role-restricted dashboard allowing the Administrator to maintain the master data that powers the entire platform.

Priority: High

ID	Requirement
FR-39	The system shall restrict access to the Admin Panel to users with the Administrator role only.
FR-40	The system shall allow the Admin to create, edit, and delete festival records (all attributes described in Section 3.3).
FR-41	The system shall allow the Admin to manage state and category master lists.
FR-42	The system shall allow the Admin to create, edit, and delete quiz questions and answer options.
FR-43	The system shall allow the Admin to view basic usage statistics (e.g., number of registered users, most-viewed festivals).

4. External Interface Requirements

4.1 User Interfaces

- A responsive, mobile-first web interface with a consistent navigation bar (Home, Explore by State, Explore by Month, Categories, Map, Calendar, Quiz, Favorites, Login/Profile).
- A homepage featuring a hero banner, a search bar, a festival-of-the-day highlight, and quick links to the Map and Calendar.
- Festival listing pages using a card-based grid layout with images, titles, and short descriptions.
- A festival detail page organized into clearly labeled sections: Overview, History, Rituals, Food, Attire, and Gallery.
- Accessible design following basic WCAG guidelines — readable font sizes, sufficient color contrast, and alt text for images.

4.2 Hardware Interfaces

The system does not directly interface with any specialized hardware. It requires only a standard client device (desktop, laptop, tablet, or smartphone) capable of running a modern web browser, and a server/host machine capable of running a Node.js application and a database instance.

4.3 Software Interfaces

Interface	Purpose
Web Browser (Client)	Renders the frontend application (HTML/CSS/JavaScript).
REST API (Backend)	Exposes endpoints consumed by the frontend for all data operations (festivals, users, quizzes, favorites).
Database System	Persists all application data — users, festivals, categories, quizzes, and favorites.
Map Library (e.g., react-simple-maps / SVG India map)	Renders the interactive, clickable India map.
Calendar Library (e.g., FullCalendar / react-calendar)	Renders the monthly festival calendar view.
Authentication Library (JWT / bcrypt)	Handles secure password hashing and session/token-based authentication.

4.4 Communication Interfaces

- All communication between the frontend and backend shall occur over HTTPS using JSON-formatted REST API requests and responses.
- The system shall use standard HTTP status codes (200, 201, 400, 401, 403, 404, 500) to indicate the result of each API call.

5. Non-Functional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-01	A festival listing page shall load within 3 seconds under normal conditions (broadband/4G, low concurrent load typical of an academic demo).
NFR-02	The interactive map shall respond to a state click/tap within 1 second.
NFR-03	The system shall support at least 50 concurrent users without noticeable degradation, sufficient for classroom/demo evaluation.

5.2 Security Requirements

ID	Requirement
NFR-04	User passwords shall be stored using a one-way hashing algorithm (e.g., bcrypt); plain-text passwords shall never be stored.
NFR-05	Authenticated routes (profile, favorites, admin panel) shall be protected using token-based authentication (JWT) or equivalent session management.
NFR-06	Admin-only functionality shall be inaccessible to Visitor and Registered User roles, enforced on both frontend and backend.
NFR-07	All user input shall be validated and sanitized on the server side to prevent injection attacks.

5.3 Usability Requirements

ID	Requirement
NFR-08	The interface shall be intuitive enough for a first-time user to locate a festival within three clicks from the homepage.
NFR-09	The system shall provide clear error and success messages for all user actions (e.g., login failure, successful bookmark).
NFR-10	The system shall be fully usable on screen widths from 360px (mobile) to 1920px (desktop).

5.4 Reliability and Availability

ID	Requirement
NFR-11	The system shall target an uptime of 99% during the evaluation/demo period, excluding scheduled maintenance.
NFR-12	The system shall handle invalid or missing data gracefully, without crashing, by displaying an appropriate fallback message.

5.5 Scalability

The system's layered architecture (see Section 6) shall allow the festival database to grow from an initial demo dataset (approximately 40–60 festivals) to a much larger catalog without requiring redesign of the frontend or API contracts.

5.6 Maintainability

- The codebase shall follow a modular folder structure separating routes, controllers, models, and UI components.
- Code shall be commented and follow a consistent naming convention to support future enhancement by new student contributors.

5.7 Portability

The system shall run on any operating system capable of hosting Node.js and a modern browser (Windows, macOS, Linux), with no OS-specific dependencies.

6. System Architecture

6.1 Architectural Overview

Festival Explorer follows a standard three-tier web architecture, which cleanly separates concerns and is well suited to a student project team divided into frontend, backend, and database sub-teams.

```
+----------------------------------------------+
|              PRESENTATION LAYER              |
|   React.js Web Client (Pages, Components)    |
+----------------------------------------------+
                          |
             HTTPS / REST (JSON)
                          v
+----------------------------------------------+
|              APPLICATION LAYER               |
|  Node.js + Express.js (Auth, Festival, Quiz, |
|              Admin APIs)                     |
+----------------------------------------------+
                          |
                 Queries / ODM
                          v
+----------------------------------------------+
|                 DATA LAYER                   |
|   MongoDB / MySQL (Users, Festivals, etc.)   |
+----------------------------------------------+
```

Figure 6.1 – Three-Tier System Architecture

6.2 Recommended Technology Stack

Layer	Technology (Recommended)	Notes
Frontend	React.js, HTML5, CSS3, JavaScript (ES6+)	Component-based UI; Tailwind CSS or Bootstrap for styling
Map Component	react-simple-maps / topojson (India states)	Renders interactive SVG map
Calendar Component	react-calendar / FullCalendar	Renders monthly festival calendar
Backend	Node.js with Express.js	REST API layer implementing business logic
Authentication	JWT + bcrypt	Secure, stateless authentication
Database	MongoDB (Mongoose ODM) or MySQL	Stores users, festivals, categories, quizzes, favorites
Hosting	Vercel/Netlify (frontend), Render/Railway (backend)	Free-tier friendly for academic deployment
Version Control	Git & GitHub	Source control and team collaboration

6.3 High-Level Module Breakdown

- Auth Module – registration, login, JWT issuance, password reset.
- Festival Module – CRUD for festival data; browsing, search, and filter endpoints.
- Map & Calendar Module – endpoints supplying state-aggregated and date-aggregated festival data.
- Recommendation Module – preference-matching logic to generate suggested festivals.
- Quiz Module – quiz question bank, submission handling, and scoring.
- Favorites Module – add/remove/list bookmarked festivals per user.
- Admin Module – secured CRUD interfaces for festivals, categories, and quizzes.

7. Data Model

7.1 Entity-Relationship Overview

The core entities of the system and their relationships are summarized below. A registered User can have many Favorites and many QuizAttempts. A Festival belongs to one State and one Category, and can appear in many Favorites and be referenced by many QuizQuestions.

```
USER (1) -----< FAVORITE >----- (1) FESTIVAL
     |                                  |
  (1)|                             (M) |(1)
     v                                  v
QUIZ_ATTEMPT                   STATE -- CATEGORY
                       |(M)
                        v
                  QUIZ_QUESTION
```

Figure 7.1 – Simplified Entity Relationship Diagram

7.2 Database Schema (Logical Design)

7.2.1 User

Field	Type	Description
userId	ObjectId / INT (PK)	Unique identifier
name	String	Full name of the user
email	String (Unique)	Login identifier
passwordHash	String	Bcrypt-hashed password
role	Enum (visitor, user, admin)	Access control role
preferredStates	Array<String>	Used by recommendation module
preferredCategories	Array<String>	Used by recommendation module
createdAt	Date	Account creation timestamp

7.2.2 Festival

Field	Type	Description
festivalId	ObjectId / INT (PK)	Unique identifier
name	String	Festival name
stateId	FK → State	State(s) where celebrated
categoryId	FK → Category	Category classification
month	String / Enum	Month(s) of celebration

Field	Type	Description
history	Text	Historical background
significance	Text	Cultural significance
rituals	Text	Rituals and traditions
food	Text	Traditional food
attire	Text	Traditional attire
images	Array<String>	Image URLs for gallery

7.2.3 State, Category, Favorite, Quiz (Summary)

Entity	Key Fields
State	stateId (PK), name, region, festivalCount
Category	categoryId (PK), name, description
Favorite	favoriteId (PK), userId (FK), festivalId (FK), addedOn
QuizQuestion	questionId (PK), festivalId/categoryId (FK), questionText, options[], correctOption, explanation
QuizAttempt	attemptId (PK), userId (FK), quizId, score, attemptedOn

8. Use Case Model

8.1 Actors

- Visitor – an unauthenticated user of the public site.
- Registered User – an authenticated user with a personal profile.
- Administrator – a privileged user who manages platform content.

8.2 Use Case Summary Table

Use Case ID	Use Case Name	Primary Actor(s)
UC-01	Register Account	Visitor
UC-02	Login / Logout	Registered User
UC-03	Browse Festivals by State/Month/Category	Visitor, Registered User
UC-04	View Festival Details	Visitor, Registered User
UC-05	Explore Interactive Map	Visitor, Registered User
UC-06	View Festival Calendar	Visitor, Registered User
UC-07	Set Preferences & View Recommendations	Registered User
UC-08	Attempt Cultural Quiz	Visitor, Registered User
UC-09	Search Festivals	Visitor, Registered User
UC-10	Manage Favorites	Registered User
UC-11	Manage Festival Content	Administrator
UC-12	Manage Quiz Questions	Administrator

8.3 Detailed Use Case: UC-04 – View Festival Details

Field	Description
Use Case ID / Name	UC-04 / View Festival Details
Actors	Visitor, Registered User
Preconditions	The user is on a festival listing, search results, map, or calendar view.
Main Flow	1. User selects a festival card or marker. 2. System retrieves festival data from the Festival Module. 3. System displays the detail page with history, rituals, food, attire, and gallery. 4. (If logged in) System shows a Favorite/Unfavorite toggle button.
Alternate Flow	If the festival record is not found, the system displays a “Festival not found” message and a link back to Explore.
Postconditions	Festival information is displayed to the user.

8.4 Detailed Use Case: UC-07 – Set Preferences & View Recommendations

Field	Description
Use Case ID / Name	UC-07 / Set Preferences & View Recommendations
Actors	Registered User
Preconditions	User is logged in.
Main Flow	1. User navigates to Profile > Preferences. 2. User selects preferred states and/or categories. 3. System saves preferences to the User record. 4. System computes a ranked list of recommended festivals by matching preferences against the Festival catalog. 5. System displays the “Recommended for You” section on the Home/Dashboard page.
Alternate Flow	If no preferences are set, the system displays trending/popular festivals as a default recommendation set.
Postconditions	User sees a personalized list of recommended festivals.

9. Requirements Traceability Matrix

The traceability matrix below maps each system feature to its related use case(s) and test focus area, ensuring every requirement can be verified during testing and demonstrated during evaluation.

Feature (Section)	Requirement IDs	Related Use Case(s)	Test Focus
3.1 Registration & Login	FR-01–FR-07	UC-01, UC-02	Auth flows, validation, security
3.2 Browse Festivals	FR-08–FR-12	UC-03	Filter correctness, combined filters
3.3 Festival Detail Page	FR-13–FR-19	UC-04	Data completeness, favorite toggle
3.4 Interactive Map	FR-20–FR-23	UC-05	Click accuracy, responsiveness
3.5 Festival Calendar	FR-24–FR-26	UC-06	Date mapping accuracy
3.6 Recommendations	FR-27–FR-29	UC-07	Preference matching logic
3.7 Quiz Module	FR-30–FR-33	UC-08	Scoring accuracy, history storage
3.8 Search & Filter	FR-34–FR-36	UC-09	Search relevance, empty state
3.9 Favorites	FR-37–FR-38	UC-10	Add/remove persistence
3.10 Admin Panel	FR-39–FR-43	UC-11, UC-12	Role restriction, CRUD correctness

10. Testing Approach and Acceptance Criteria

10.1 Testing Levels

Test Level	Description	Example
Unit Testing	Verify individual functions/components in isolation.	Password hashing function returns a valid hash.
Integration Testing	Verify interaction between frontend, API, and database.	Login API correctly issues a JWT consumed by the frontend.
System Testing	Verify the complete application against functional requirements.	End-to-end: browse → view details → add favorite.
User Acceptance Testing (UAT)	Faculty/peers validate the system against expected outcomes.	Evaluator confirms recommendations reflect chosen preferences.

10.2 Sample Acceptance Criteria

- Given a valid email and password, when a user submits the login form, the system shall authenticate and redirect to the homepage within 2 seconds.
- Given a state is selected on the map, when the click event fires, the system shall display the correct list of festivals associated with that state only.
- Given a user completes a quiz, when they submit their answers, the system shall display an accurate score matching the number of correct responses.
- Given an Admin deletes a festival, when the deletion is confirmed, the festival shall no longer appear in any listing, search result, or the map.

11. Appendices

Appendix A – Proposed Screen List

- Home / Landing Page
- Explore by State Page
- Explore by Month Page
- Explore by Category Page
- Festival Detail Page
- Interactive India Map Page
- Festival Calendar Page
- Search Results Page
- Quiz Landing & Quiz Attempt Page
- Quiz Result Page
- Login / Register Page
- User Profile & Preferences Page
- My Favorites Page
- Admin Dashboard
- Admin – Manage Festivals / Categories / Quiz

Appendix B – Future Enhancements (Out of Current Scope)

- Multi-language support (Hindi and other regional languages).
- AI/ML-based recommendation engine using collaborative filtering.
- User-generated content — photo submissions and festival stories with moderation.
- Integration with a public events/ticketing API for festival-related events.
- Progressive Web App (PWA) / native mobile app version.
- Gamification — badges and leaderboards for quiz performance.

Appendix C – Glossary

Term	Meaning
Card-based UI	A design pattern displaying each item (festival) as a compact rectangular “card” with an image and summary text.
Rules-based Recommendation	A recommendation approach using explicit matching rules rather than a trained machine-learning model.
Role-Based Access Control (RBAC)	A security model restricting system functions based on the user's assigned role.
Responsive Web Design	A design technique enabling a single web layout to adapt to different screen sizes.

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